Sachi Patel

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EDUCATION

Barnard College of Columbia University — Expected May 2027
B.A. in Computer Science & Economics | **GPA:** 3.91 | New Pathways Bridgewater Scholar — *New York, NY*

- **Selected Coursework:** Data Structures, Computer Architecture, Systems Programming, Natural Language Processing, Artificial Intelligence, Applied Machine Learning, Databases, Computer Science Theory, Discrete Math, Linear Algebra

SKILLS

Languages/Frameworks: Python, Java, JavaScript, TypeScript, C, SQL, R, MATLAB; React, Node.js, Flask, FastAPI, Next.js
ML/NLP: PyTorch, scikit-learn, HuggingFace, LangChain, FAISS, Pandas, NumPy
Data/Cloud/DevOps: PostgreSQL, BigQuery, AWS (S3, Lambda, Cost Explorer), Docker, Git, Terraform, CI/CD, REST APIs

WORK EXPERIENCE

Depository Trust and Clearing Corporation — Jun. 2025 – Aug. 2025
Software Engineering Intern — *Jersey City, NJ*

- Built internal automation to detect idle EC2/EBS volumes across 25+ SYSIDs; alerts drove $350K in annualized savings
- Developed MVP cost-savings dashboard (Python, Flask, REST APIs) integrating CloudAware and AWS Cost Explorer data
- Presented cost insights to CloudOps teams and SYSID owners; drove decommissioning of 200+ idle resources org-wide

Rajiv Sethi Lab, Barnard College — Jan. 2024 – Present
Research Assistant — *New York, NY*

- Built data pipeline (Python, BeautifulSoup) scraping price data/probabilities from prediction markets/models across 41 events
- Implemented profitability test (Python/MATLAB) analyzing trading strategies; extracted insights on model vs. market accuracy
- Co-authored paper finding FiveThirtyEight beat Polymarket by 17.9% on popular vote; published in ACM CI '25 proceedings

Martina Jasova Lab, Barnard College — Jul. 2024 – Dec. 2024
Research Assistant — *New York, NY*

- Developed text extraction pipeline (PyMuPDF, concurrent.futures) to parse and chunk 12K+ pages of financial reports
- Integrated GPT-4o mini for ESG sentiment classification with batched inference, retry logic, and exponential backoff
- Computed alignment scores (cosine similarity) between 50+ banks' ESG claims and actual sustainable lending activity

Barnard Computational Science Center (CSC) — Aug. 2024 – Present
Senior Computing Fellow — *New York, NY*

- Mentored 120+ students in R, MATLAB, Python across 3 data science courses; debugged code and guided final projects
- Led 30+ office hours sessions supporting Coding Markets, Privacy in a Networked World, and Intro to Data Science
- Adapted and led 3 workshops; designed Colab notebooks with modular exercises for data wrangling, visualization, and scripting

PROJECTS

RAG-Powered Codebase Assistant — Jan. 2026

- Built RAG assistant that ingests GitHub repos, parses Python code w/ tree-sitter, answers questions via natural language queries
- Implemented semantic search using sentence-transformers/FAISS; integrated LangChain + Ollama for conversational UI
- Engineered multi-repo indexing with chunked parsing; achieved sub-second query latency across 10K+ lines of code

LLM Factuality Evaluation for Medical Summarization — Dec. 2025

- Built QAGS-style pipeline comparing factual consistency of LLaMA-8B, Mistral-7B, DeepSeek-7B on CORD-19 papers
- Created evaluation system: automated question generation (FLAN-T5), extractive QA (RoBERTa), F1 scoring w/ BERTScore
- Optimized pipeline for 8K-token docs; findings on weak factual grounding of models (<0.50 F1) published in project papers

Agentic Stock Prediction System — Feb. 2026 (*In Progress*)

- Building autonomous stock prediction agent (Google ADK, Python) using historical ticker data and technical indicators
- Integrating LLM-based reasoning with real-time market APIs to generate actionable buy/sell/hold recommendations

Full-Stack Restaurant Reservation System — Feb. 2025

- Built full-stack app (Flask, PostgreSQL, React) with JWT auth, real-time reservations, and normalized relational schema
- Wrote 40+ unit tests (pytest) with 85% coverage; containerized with Docker/deployed on AWS EC2 via GitHub Actions

HTTP Web Server — Dec. 2024

- Built HTTP/1.0 server in C using POSIX sockets; forked child processes to handle concurrent connections and serve static files
- Implemented request parsing, URI validation, directory traversal prevention, and error handling (200/301/400/404/501)

ACTIVITIES & LEADERSHIP

Columbia Multicultural Business Association (Conference Lead) | Society of Women Engineers (Committee Member) | Columbia Women in Computer Science (Event Coordinator) | Grace Hopper Conference '24 & '25 | Columbia DJ Collective